Solution Requirements

Wandr AI – AI Travel Itinerary Planner

1. Functional Requirements

ID	Requirement	Priority	Source
FR-01	User shall register with email and password via Supabase Auth	Must	Security design
FR-02	User shall log in and receive a JWT valid for the Supabase session	Must	Security design
FR-03	User shall submit a trip generation form with destination, dates, budget, style, pace, interests	Must	Core feature
FR-04	Backend shall call Groq API and return a complete itinerary in ≤ 30 s	Must	Performance
FR-05	System shall store trips in Supabase with user_id foreign key	Must	Data persistence
FR-06	User shall view a paginated list of their saved trips	Must	Core feature
FR-07	User shall view the full day-by-day itinerary for a trip	Must	Core feature
FR-08	User shall edit trip metadata (title, status, budget, notes)	Should	Core feature
FR-09	User shall regenerate a single day without affecting other days	Should	Differentiator
FR-10	User shall view a budget breakdown with per-day costs	Should	Core feature
FR-11	User shall delete a trip	Must	Data ownership

FR-12	System shall return correct CORS headers for approved origins	Must	Security
FR-13	System shall reject unauthorised API requests with HTTP 401	Must	Security

2. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	API responses (excluding AI generation) < 500 ms at p95
NFR-02	AI Latency	Groq generation completes in < 30 s; timeout at 120 s
NFR-03	Availability	Backend uptime ≥ 99% (Render paid) or best-effort free tier
NFR-04	Security	All secrets (GROQ_API_KEY, SUPABASE_JWT_SECRET, service_role key) stored in Render env vars, never in source code
NFR-05	Data Isolation	Supabase RLS ensures user_id-scoped data access at DB level
NFR-06	Scalability	Stateless FastAPI + Supabase horizontal scaling; Groq rate-limiting handled by tenacity retry
NFR-07	Usability	Trip creation form completable in under 2 minutes
NFR-08	Reliability	Groq API retried up to 3 times with exponential backoff on timeout/connection errors